MUTUAL DIVISIBILITY IN IWASAWA ALGEBRAS AND THE PRINCIPAL-IDEAL PINCH

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Abstract. We prove that mutual divisibility of two nonzero elements in a unique factorization domain (UFD) forces equality of the generated principal ideals. The proof is a short argument from the uniqueness of irreducible factorization. Specializing to the Iwasawa algebra $\Lambda = \mathbb{Z}_p[\![T]\!]$ (a UFD by the p-adic Weierstrass preparation theorem), we obtain: if $A \mid B$ and $B \mid A$ in Λ, then $(A) = (B)$. This *principal-ideal pinch* is the purely algebraic step that upgrades one-sided Iwasawa-theoretic divisibility results (such as Kato's divisibility and a reverse divisibility from Λ-adic height positivity) into the full cyclotomic Iwasawa Main Conjecture equality $\mathrm{char}_\Lambda X_p = (L_p)$ for each fixed good prime p. We also record a finite-to-infinite capacity obstruction used in companion papers for topological veto arguments.

1. Mutual divisibility in UFDs

The core algebraic observation is entirely elementary.

Lemma 1.1 (Mutual divisibility forces association)**.** *Let R be a unique factorization domain and $A, B \in R \setminus \{0\}$. If $A \mid B$ and $B \mid A$, then A and B are associates: there exists a unit $u \in R^\times$ with $A = uB$.*

Proof. Since R is a UFD, write

$$A = \pi \prod_{i \in I} P_i^{a_i}, \qquad B = \pi' \prod_{i \in I} P_i^{b_i},$$

where $\pi, \pi' \in R^\times$ are units, I is a finite index set, the P_i are pairwise non-associate irreducible elements, and $a_i, b_i \geq 0$.

The divisibility $A \mid B$ means $B = A \cdot c$ for some $c \in R$; comparing irreducible factorizations (which are unique up to units and reordering in a UFD) gives $a_i \leq b_i$ for all $i \in I$.

Similarly, $B \mid A$ gives $b_i \leq a_i$ for all $i \in I$.

Hence $a_i = b_i$ for every i. Therefore

$$\frac{A}{B} = \frac{\pi}{\pi'} \in R^\times,$$

so $A = (\pi/\pi')B$ with π/π' a unit. □

Corollary 1.2 (Principal-ideal equality)**.** *Under the hypotheses of Lemma 1.1, $(A) = (B)$ as ideals of R.*

Proof. $A = uB$ with $u \in R^\times$ gives $(A) \subset (B)$. $B = u^{-1}A$ gives $(B) \subset (A)$. Hence $(A) = (B)$. □

Remark 1.3 (Necessity of UFD). The UFD hypothesis is essential. In the ring $\mathbb{Z}[\sqrt{-5}]$ (which is not a UFD), we have $2 \mid 6$ and $3 \mid 6$, but $(2) \neq (3)$. More to the point, mutual divisibility can fail to imply association in non-UFD domains; see [8, Chapter 1] for examples.

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2. Application to Iwasawa algebras

We now specialize to the ring central to Iwasawa theory.

Definition 2.1 (Iwasawa algebra)**.** For a prime p, the *Iwasawa algebra* is the formal power series ring $\Lambda := \mathbb{Z}_p[\![T]\!]$, where $\mathbb{Z}_p$ denotes the p-adic integers.

Proposition 2.2 (Λ is a UFD)**.** *$\Lambda = \mathbb{Z}_p[\![T]\!]$ is a unique factorization domain.*

Proof. By the p-adic Weierstrass preparation theorem (see Washington [9, §7.1] or Neukirch–Schmidt–Wingberg [7, §V.1]), every nonzero $f \in \Lambda$ factors uniquely as

$$f = p^a \cdot g(T) \cdot u(T),$$

where $a \geq 0$ is an integer, $g(T) \in \mathbb{Z}_p[T]$ is a distinguished (monic, with non-leading coefficients in $p\mathbb{Z}_p$) polynomial, and $u(T) \in \Lambda^\times$ is a unit power series. The element p is irreducible in Λ. Distinguished polynomials factor into irreducible distinguished polynomials (using the fact that $\mathbb{Z}_p[T]$ is a UFD, by Gauss's lemma applied to the DVR $\mathbb{Z}_p$). Thus Λ has unique factorization into p and irreducible distinguished polynomials, up to units. □

Theorem 2.3 (Principal-ideal pinch in Λ)**.** *Let $A, B \in \Lambda \setminus \{0\}$ with $A \mid B$ and $B \mid A$. Then $(A) = (B)$ in Λ.*

Proof. Λ is a UFD by Proposition 2.2. Apply Lemma 1.1 and Corollary 1.2. □

Corollary 2.4 (IMC equality from two-sided divisibility)**.** *Let $E/\mathbb{Q}$ be an elliptic curve with good reduction at a prime $p \geq 5$, and assume E is modular (by Wiles et al.). Let $\xi_p(T) \in \Lambda$ be a generator of the characteristic ideal $\operatorname{char}_\Lambda X_p(E/\mathbb{Q}_\infty)$ (where X_p is the Pontryagin dual of the p-primary Selmer group over the cyclotomic $\mathbb{Z}_p$-extension), and let $L_p(E,T) \in \Lambda$ be the cyclotomic p-adic L-function (Mazur–Swinnerton-Dyer). If*

(a) *$\xi_p \mid L_p$ in Λ (Kato's one-sided divisibility [4]), and*
(b) *$L_p \mid \xi_p$ in Λ (reverse divisibility from Λ-adic height positivity [1]),*

then $\operatorname{char}_\Lambda X_p = (L_p)$ as ideals of Λ (equality up to $\Lambda^\times$).

Proof. Apply Theorem 2.3 with $A = \xi_p$ and $B = L_p$. Hypotheses (a) and (b) give $A \mid B$ and $B \mid A$. Theorem 2.3 yields $(A) = (B)$, i.e. $\operatorname{char}_\Lambda X_p = (\xi_p) = (L_p)$. □

Remark 2.5 (Sources of the two divisibilities)*.* Condition (a) is the celebrated result of Kato [4] for ordinary primes, with supersingular extensions by Kobayashi [5] (signed Selmer groups) and Lei–Loeffler–Zerbes [6] (Wach modules). Condition (b) is the reverse divisibility established in the companion BSD paper [1] via Λ-adic Néron–Tate height positivity. Corollary 2.4 shows that the algebraic step from two one-sided divisibilities to full Main Conjecture equality is a single application of unique factorization—a one-line argument once the UFD property is in hand.

3. Finite-to-infinite capacity obstruction

We record a combinatorial observation used in companion papers.

Proposition 3.1 (Finite cannot surject onto infinite)**.** *If S is a finite set and T is an infinite set, there is no surjection $f : S \to T$.*

Proof. If $f : S \to T$ is a surjection, then $|T| \leq |f(S)| \leq |S| < \infty$, contradicting $|T| = \infty$. □

Corollary 3.2 (Finite-budget obstruction)**.** *If each operation costs $c > 0$ and the total budget is $B < \infty$, then at most $\lfloor B/c \rfloor$ operations can be performed. In particular, infinitely many operations require infinite budget.*

Proof. n operations cost nc. The constraint $nc \leq B$ gives $n \leq B/c$, hence $n \leq \lfloor B/c \rfloor$. The contrapositive: if infinitely many operations are needed, the cost is $\sum_{n=1}^{\infty} c = \infty > B$. □

Remark 3.3 (Application to topological veto). This elementary budget argument is the algebraic backbone of the topological-capacity veto in [3]: when each vortex-line crossing incurs cost $\ln \varphi > 0$ (the link penalty from [2]) and the initial data has finite H^1 energy, only finitely many crossings can occur—ruling out blow-up profiles that would require infinitely many.

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